

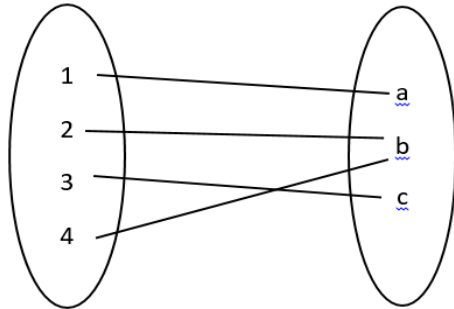
UNIT-1

1. Explain the concept of a set with suitable examples and discuss various types of sets such as universal set, subset, empty set, equal sets, cardinality of a set, power set, and Cartesian product.
2. Using Venn diagrams, show the union, intersection, and set difference of two given sets.
3. Explain the concept of Cartesian product and find $A \times B$ given $A = \{1, 2\}$ and $B = \{a, b, c\}$.
4. Explain the various set operations with suitable examples, including union, intersection, disjoint sets, complement, set difference, and symmetric difference.
5. Explain and justify the Principle of Inclusion and Exclusion with an appropriate example.
6. Using a Venn diagram, show the relationship between the sets $A = \{2, 3, 5, 7\}$ and $B = \{5, 7, 9, 10\}$.
7. Verify the logical equivalence of $\sim(p \wedge q) \equiv \sim p \vee \sim q$.
8. $(p \vee q) \rightarrow r \equiv (p \rightarrow r) \wedge (q \rightarrow r)$ state whether it is logically equivalent or not.
9. Is $Q \vee (P \wedge \neg Q) \vee (\neg P \wedge \neg Q)$ a tautology? Explain by using truth table.
10. What are the rules for constructing well-formed formulas in statement calculus?
11. Demonstrate by Mathematical Induction that $P(n) = 1 + 2 + 3 + \dots + n = \frac{n(n+1)}{2}$ for any $n \geq 1$.
12. Using Mathematical Induction, show that $1 + 2 + 2^2 + \dots + 2^n = 2^{n+1} - 1$ for any $n \geq 1$.
13. Apply the method of Mathematical Induction to prove that $1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n}{6}(n+1)(2n+1)$, $n \geq 1$.
14. Show that every non-empty subset of positive integers contains a smallest element.
15. Using Mathematical Induction, show that $1 + 2 + 2^2 + \dots + 2^n = 2^{n+1} - 1$ for any $n \geq 1$.
16. Prove by mathematical Induction that $1^2 + 2^2 + 3^2 + \dots + n^2 = \frac{n}{6}(n+1)(2n+1)$, $n \geq 1$.
17. Show that every non-empty set of natural numbers has a least element.
18. List the steps involved in proving Euclid's Algorithm.
19. Find the GCD of :i) (12,33)
ii) (750,900)
iii) (252,105)
iv) (1005,105)
v) (48,18)
vi) (56,36)

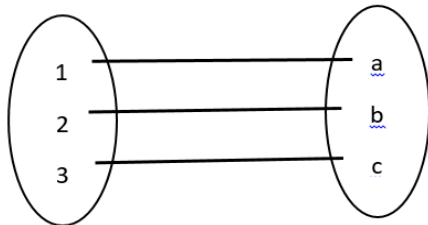
UNIT-2

1. Given the set $A = \{1, 2, 3\}$ and the relation $R = \{(1, 1), (2, 2), (3, 3), (1, 2), (2, 1)\}$ show that R is an equivalence relation on A .
2. Verify the reflexive property of the relation $a \leq b \wedge b \leq a$ on integers.
3. State the definition of a reflexive relation along with one example.
4. Explain the concept of a symmetric relation with a suitable example.
5. Define a transitive relation and illustrate it with an example.
6. What do you understand by an equivalence relation? Explain it with an example.

7. Check if the relation $R = \{(1,2), (2,1)\}$ satisfies the condition of symmetry on $A = \{1,2\}$.
8. Let $A = \{1,2,3,4\}$ and $B = \{a,b,c\}$, $R = \{(1,a), (1,b), (2,b), (2,c), (3,b), (4,a)\}$
- $S = \{(1,b), (2,c), (3,b), (4,b)\}$
 - Compute (a) $\bar{R}$ (b) $R \cap S$ (c) $R \cup S$ (d) R^{-1}
9. Let $A = \{1,2,3,4\}$ let R be the relation defined on A as $a R b$ if and only if $a < b$.
10. Identify the types $A = \{1,2,3,4\}$ $B = \{a,b,c\}$ $f = \{(1,a), (2,b), (3,c), (4,b)\}$



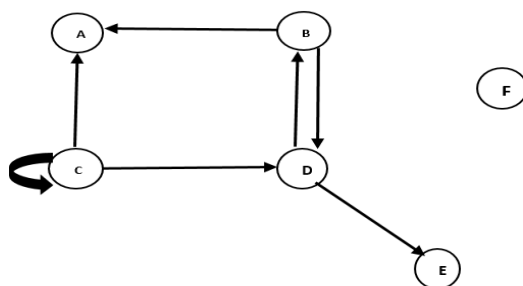
11. $A = \{1,2,3\}$ $B = \{a,b,c\}$ $F = \{(1,a), (2,b), (3,c)\}$, Identify the type.



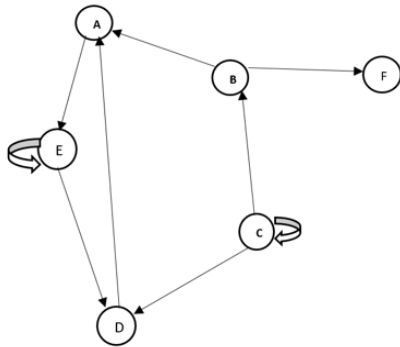
12. Let $A = \{1,2,3,4\}$, $B = \{a,b,c\}$ $f = \{(1,a), (2,b), (3,c), (4,b)\}$ **Identify the type of function.**
13. Let $A = \{1,2,3\}$, $B = \{a,b,c,d\}$ $f = \{(1,a), (2,b), (3,c)\}$ **Identify the type of function.**
14. Let $A = \{p,q,r\}$, $B = \{1,2\}$ $f = \{(p,1), (q,2), (r,1)\}$ **Identify the type of function.**
15. Let $A = \{x,y,z\}$, $B = \{1,2,3\}$ $f = \{(x,1), (y,2), (z,2)\}$ **Identify the type of function.**

UNIT 3 QB

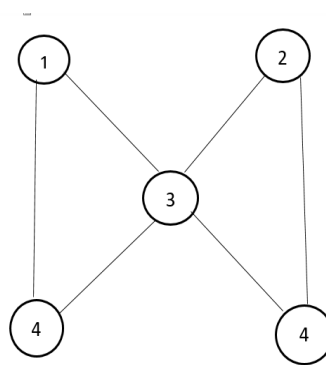
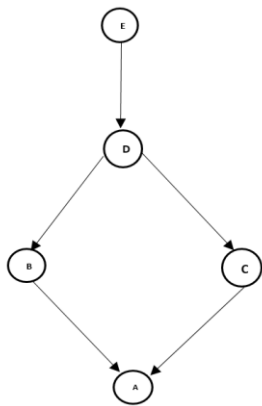
- Explain the concepts of multigraph, weighted graph, and Euler graph using appropriate diagrams.
- Evaluate the indegree and outdegree of every vertex in the given directed graph.



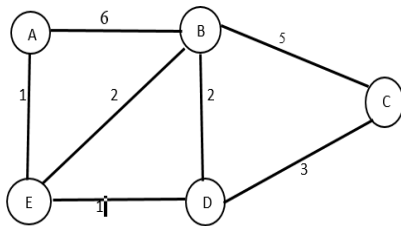
3. Using the given graph, obtain the indegree and outdegree of its vertices.



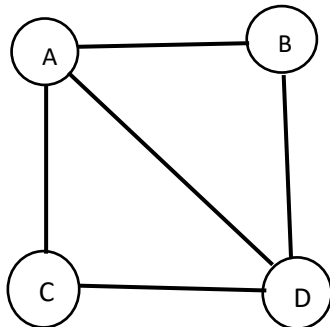
4. Explain the concepts of Euler path and Euler circuit with the help of graphs.
 5. Check if the given graph contains an Euler path or an Euler circuit.



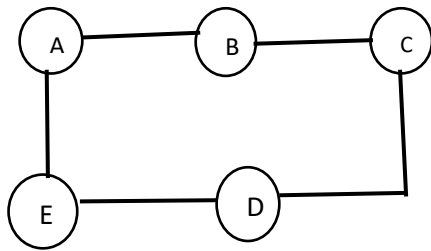
6. Explain the concept of a circuit in a graph with the help of a diagram.
 7. State the meaning of a path in a graph and illustrate it graphically.
 8. Use Dijkstra's algorithm to determine the shortest path.



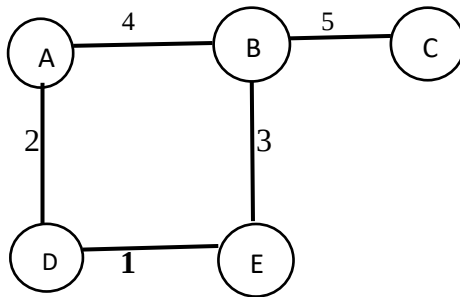
9. Explain the concept of a circuit in a graph with the help of the above figure.



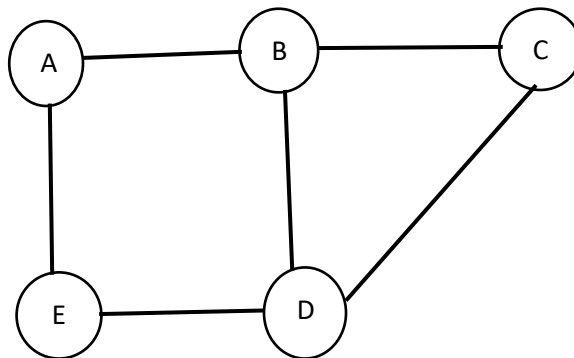
10. Check whether the graph contains an Euler path. Give reason.



11. Use Dijkstra's algorithm to find the shortest path from vertex A to C.



12. Explain Euler path and Euler circuit. Check whether the given graph contains an Euler path or an Euler circuit.



UNIT 4 QB

1. Demonstrate that the set $G = \{1, -1, i, -i\}$ is a group with respect to multiplication.
2. State the definitions of a group, semigroup, and normal subgroup in algebra.
3. A subgroup H of a group G is normal iff $x H x^{-1} = H \quad \forall x \in G$
4. Show that the matrices $A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ $B = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}$

$$C = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad D = \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$$

5. Show that the set of integers forms a group under addition.

6. Show that $M_2(\mathbb{R})$ is a group with respect to addition.

7. A subgroup H of a group G is normal iff $g^{-1} h g \in H$, for every $h \in H, g \in G$.

8. Define Monoid , sub-group.